



Decision Theory &

Objective of Classification Models

Inductive Learning Hypothesis : Interpretation

- Target Concept : t
- Discrete : $f(x) \in \{\text{Yes, No, Maybe}\}$ Classification
- Continuous : $f(x) \in [20-100]$ Regression
- Probability Estimation : $f(x) \in [0-1]$

Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport?
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

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Decision Theory

- Target Concept : t
- Discrete : $f(x) \in \{\text{Yes, No}\}$ ie., $t \in \{0, 1\}$ Binary Classification
- Continuous : $f(x) \in [20-100]$
- Probability Estimation : $f(x) \in [0-1]$

ML Task : Predict the Employability of interview candidates based on CGPA & IQ

Preprocess Implemented:

Min-Max Normalization on IQ

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0.5

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Decision Theory :

The decision problem: given x , predict t according to a probabilistic model $p(x, t)$

- Target Concept : t
- Discrete : $f(x) \in \{\text{Yes, No}\}$ ie., $t \in \{0, 1\}$
- Continuous : $f(x) \in [20-100]$
- Probability Estimation : $f(x) \in [0-1]$

$p(x, C_k)$ is the (central!) inference problem

CGPA	IQ	IQ	Job Offered	P(Job = 1)
5.5	6.7	100	1	0.8
5	7	105	0	0.4
8	6	90	1	0.75
9	7	105	1	0.95
6	8	120	0	0.35
7.5	7.3	110	0	0.4

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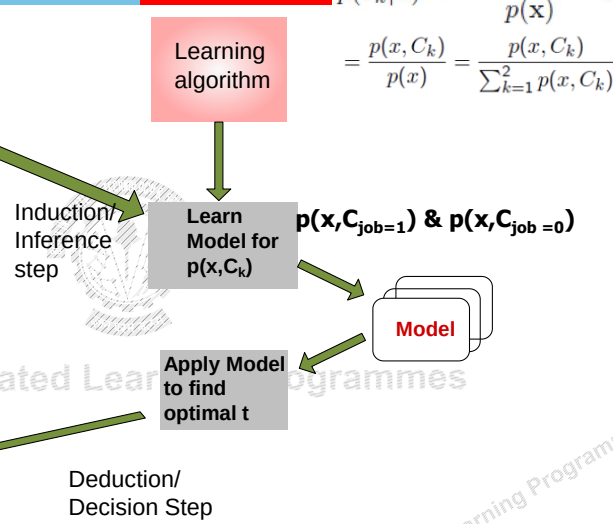
Classification Problem: Stages

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0

Training Set

CPGA	IQ	Job - Offered
3	4	?
7	6	?
5.5	8	?

Test Set



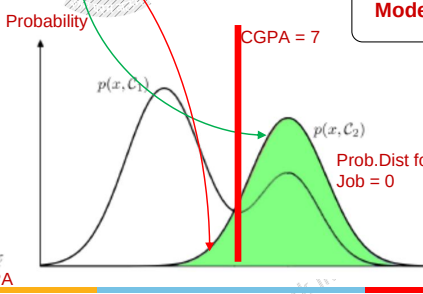
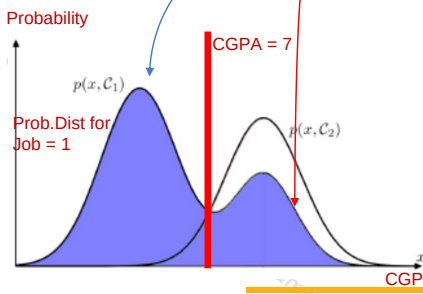
$$p(C_k|x) = \frac{p(x|C_k)p(C_k)}{p(x)}$$
$$= \frac{p(x, C_k)}{p(x)} = \frac{p(x, C_k)}{\sum_{k=1}^2 p(x, C_k)}$$

Decision Region

Sample Rule / Hypothesis:
IF CGPA < 7 Job = 1
Else Job = 0

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0

Training Set



Model divides the input space into regions **R_k** called **decision regions**, one for each class, such that all points in R_k are assigned to class C_k. A mistake occurs when an input vector belonging to class C₁ is assigned to class

Misclassification Rate

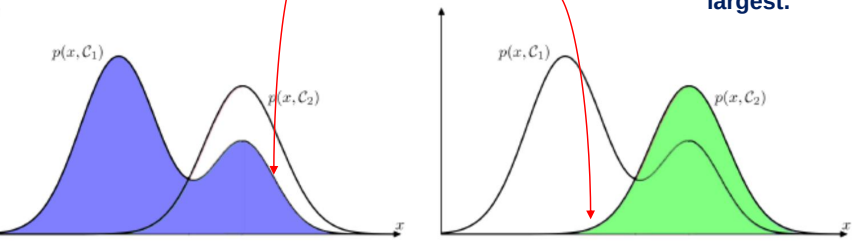
$$p(C_k|x) = \frac{p(x, C_k)}{p(x)}$$

$$p(\text{mistake}) = p(x \in R_1, C_2) + p(x \in R_2, C_1)$$
$$= \int_{R_1} p(x, C_2) dx + \int_{R_2} p(x, C_1) dx.$$

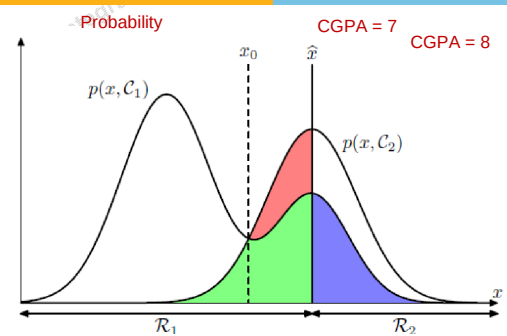
$$p(x, C_1) > p(x, C_2)$$
$$\Leftrightarrow p(C_1|x)p(x) > p(C_2|x)p(x)$$
$$\Leftrightarrow p(C_1|x) > p(C_2|x)$$

To minimize p(mistake), each x is assigned to whichever class has the smaller value of the integrand

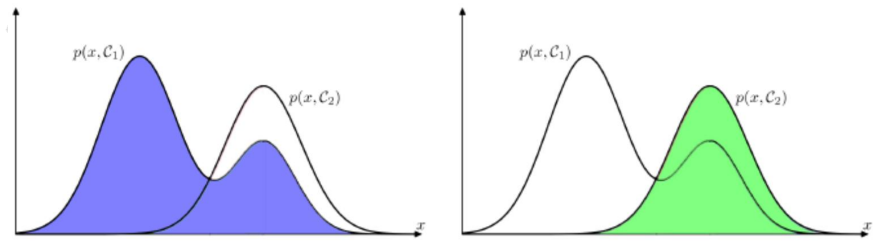
The minimum probability of making a mistake is obtained if each value of x is assigned to the class for which the **posterior probability p(C_k|x) is largest.**



Decision Theory - Summary

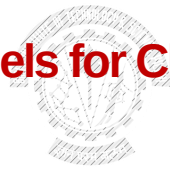


$\hat{x}$: decision boundary.
 x_0 : optimal decision boundary
 $x_0 : \arg \min_{R_1} \{p(\text{mistake})\}$





Linear Models for Classification

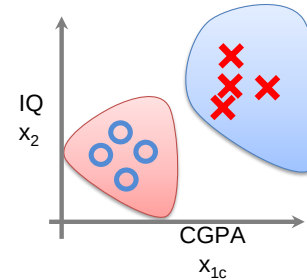


Types of Classification

Decision Theory: Interpretation

Building

Generative



$$P(Y|X_1, X_2, \dots, X_n) = \frac{P(X_1, X_2, \dots, X_n | Y) P(Y)}{P(X_1, X_2, \dots, X_n)}$$

Known as generative models, because by sampling from them it is possible to generate synthetic data points in the input space.

Eg., Classification: **Naïve Bayes**,
Clustering : **Mixtures of Gaussians**

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)}$$

Likelihood: $P(x|c)$
Class Prior Probability: $P(c)$
Posterior Probability: $P(c|x)$
Predictor Prior Probability: $P(x)$

$$P(c|X) = P(x_1|c) \times P(x_2|c) \times \dots \times P(x_n|c) \times P(c)$$

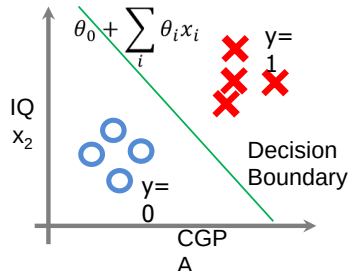
CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0
...		

Types of Classification

Decision Theory: Interpretation

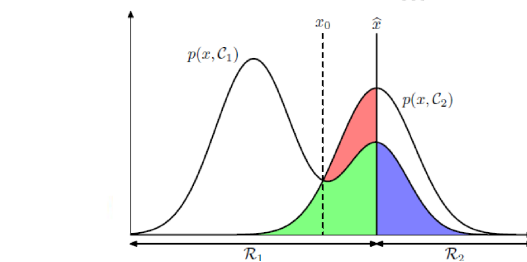
Building

Discriminative



$$\theta_0 + \sum_i \theta_i x_i \geq 0$$
$$\theta_0 + \sum_i \theta_i x_i < 0$$

Logistic regression, SVMs, tree based classifiers (e.g. decision tree) Traditional neural networks, Nearest neighbor



$$P(c|x):$$

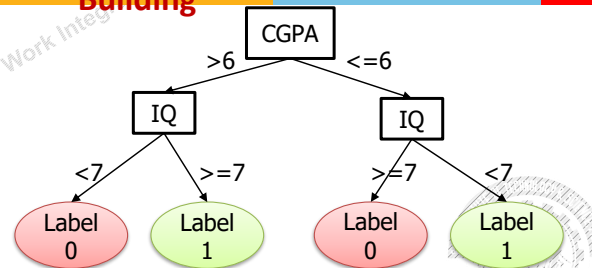
Posterior Probability

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0
...		

Types of Classification

Decision Theory: Interpretation

Building



IF CGPA > 6 and IQ ≥ 7
Job offered = 1
Else If CGPA ≤ 6 and IQ < 7
Job offered = 0
Else if CGPA ≤ 6 and IQ ≥ 7
Job offered = 1
.....

Logistic regression, SVMs, tree based classifiers (e.g. decision tree) Traditional neural networks, Nearest neighbor

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0
...		

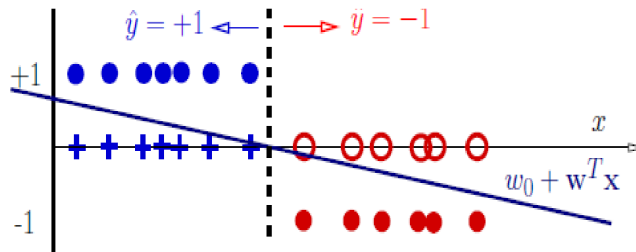
Logistic Regression



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Decision Rules



Classifier:

$f(\mathbf{x}, \mathbf{w}) = w_0 + \mathbf{w}^T \mathbf{x}$ (linear discriminant function)

$$y = \begin{cases} 1 & \text{if } f(\mathbf{x}, \mathbf{w}) \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

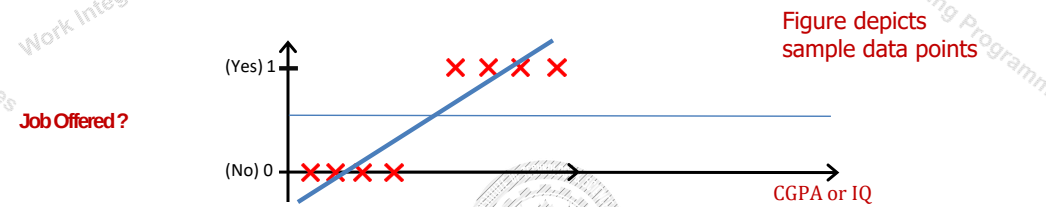
Decision rule is

Mathematically

$$y = \text{sign}(w_0 + \mathbf{w}^T \mathbf{x})$$

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Logistic Regression vs Least Squares Regression



Independent Attribute:

CGPA or IQ

A Discriminant function $f(\mathbf{x})$ directly map input to class labels
In two-class problem, $f(\cdot)$ is binary valued

If $h_{\theta}(\mathbf{x}) \geq 0.5$, predict " $y = 1$ "

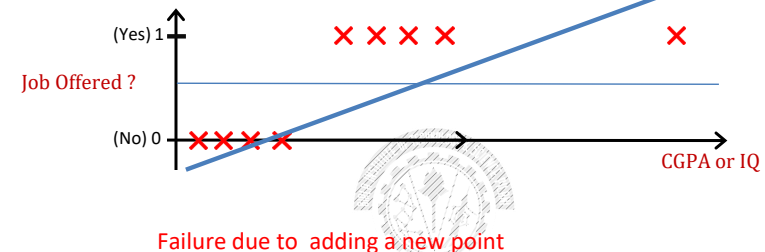
If $h_{\theta}(\mathbf{x}) < 0.5$, predict " $y = 0$ "

- Can we solve the problem using linear

In this use case : $h_{\theta}(\mathbf{x}) = 0.7$, implies that there is 70% of the chance of the candidate being selected in the interview

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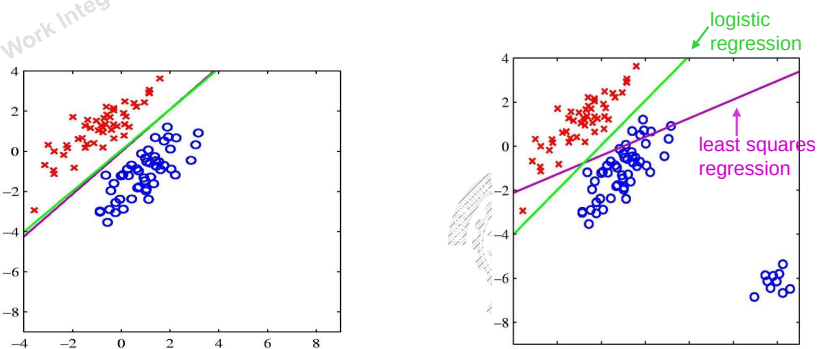
Logistic Regression vs Least Squares Regression



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Logistic Regression vs Least Squares Regression



The right-hand plot shows the corresponding results obtained when extra data points are added at the bottom left of the diagram, showing that least squares is highly sensitive to outliers, unlike logistic regression.

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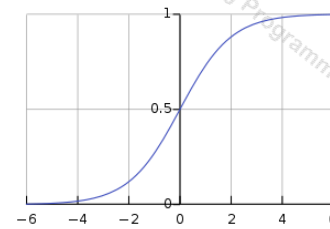
Sigmoid Function

- Sigmoid/logistic function takes a real value as input and outputs another value between 0 and 1
- That framework is called logistic regression
 - Logistic: A special mathematical sigmoid function it uses
 - Regression: Combines a weight vector with observations to create an answer

- Want $0 \leq h_{\theta}(x) \leq 1$

$$h_{\theta}(x) = g(\theta^T x),$$

$$\text{where } g(z) = \frac{1}{1+e^{-z}}$$



$$h_{\theta}(x) = g(\theta^T x)$$

- Sigmoid function

- Logistic function

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

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Logistic Regression vs Least Squares Regression

Another drawback of using linear regression for this problem

$$Y = -\infty \quad Y = +\infty$$

$$Y = \theta_0 + \sum_i \theta_i x_i$$

$$p = \frac{1}{1 + e^{-y}} \quad [h_{\theta}(x) = g(\theta^T x)]$$

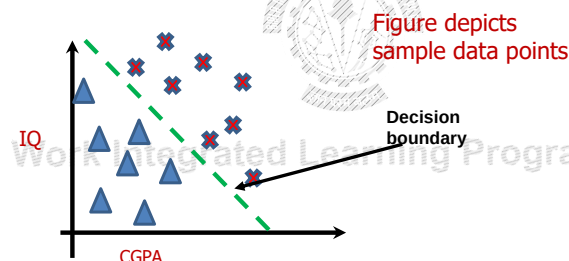
$$\ln \frac{p}{1-p} = \theta_0 + \sum_i \theta_i x_i$$

$$0 \leq h_{\theta}(x) \leq 1$$

P=0 P=1

Logistic Regression – Sample Linear Boundary

- At decision boundary output of logistic regression is 0.5
- $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$
 - e.g., $\theta_0 = -3, \theta_1 = 1, \theta_2 = 1$



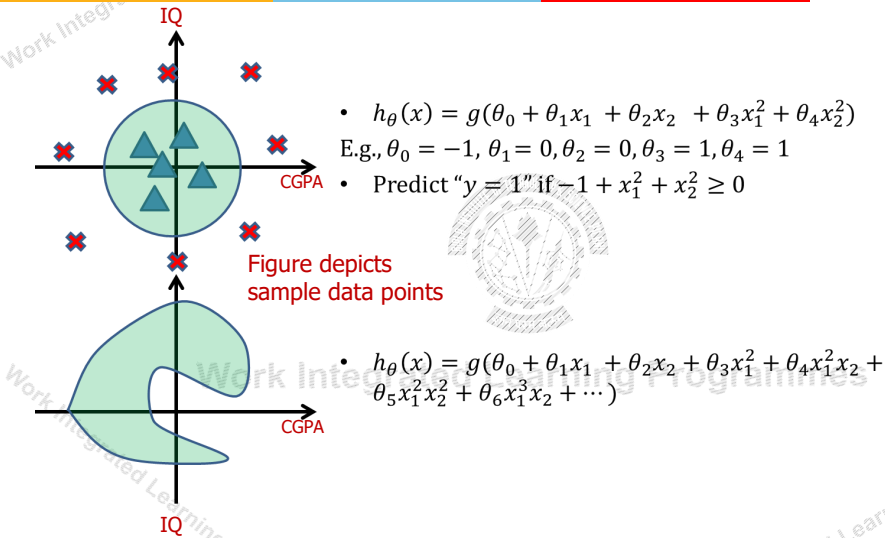
- Predict “ $y = 1$ ” if $-3 + x_1 + x_2 \geq 0$

Slide credit: Andrew Ng

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Logistic Regression – Sample Non-Linear Boundary



Slide credit: Andrew Ng

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Learning model parameters

- Training set:

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$$

- m examples

$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix} \quad x_0 = 1, y \in \{0, 1\}$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

- How to choose parameters (feature weights) ?

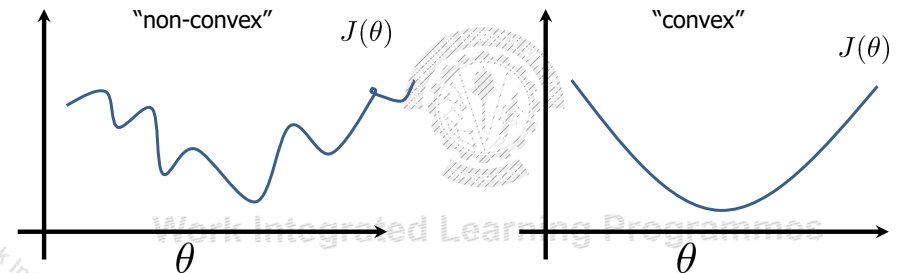
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Notion of Cost Function in Classification

Logistic Regression

- Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$
- How to choose parameters (feature weights)? θ

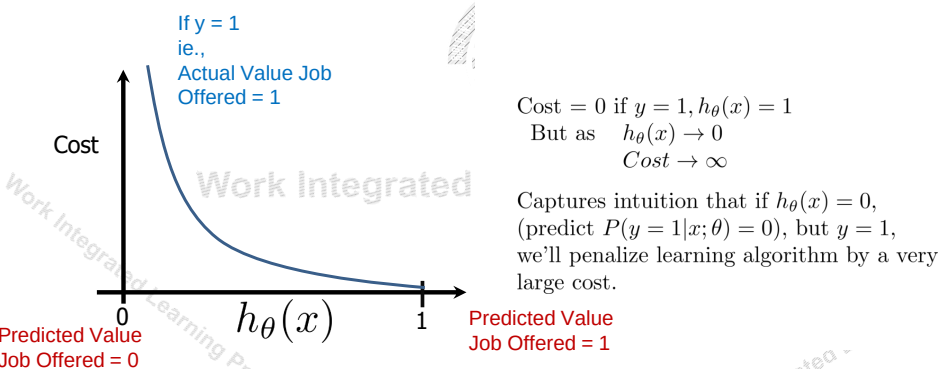


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Logistic regression cost function (cross entropy)

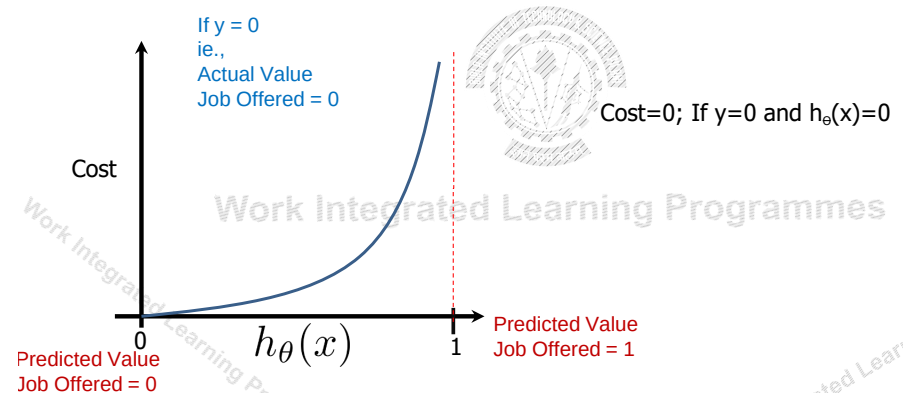
$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



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Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



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Cost function

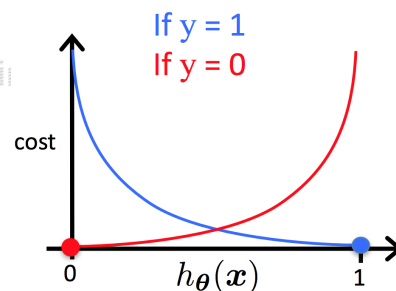
$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

To fit parameters θ : Apply Gradient Descent Algorithm $\min_{\theta} J(\theta)$

To make a prediction given new x :

Output : $h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$



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Gradient Descent Algorithm

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right]$$

Goal: $\min_{\theta} J(\theta)$

Repeat

$$\begin{cases} \theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) \end{cases}$$

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

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Gradient Descent Algorithm

Linear Regression

Repeat {
 $\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$

$$h_{\theta}(x) = \theta^T x$$

Logistic Regression

Repeat {
 $\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 CGPA + \theta_2 IQ)}}$$

Slide credit: Andrew Ng

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Application of Logistic Regression & Problem Types

Example: Sentiment Analysis – With Engineered features

It's **hokey**. There are virtually **no** surprises, and the writing is **second-rate**. So why was it so **enjoyable**? For one thing, the cast is **great**. Another **nice** touch is the music **I** was overcome with the urge to get off the couch and start dancing. It sucked **me** in, and it'll do the same to **you**.

$x_1=3$ $x_2=2$ $x_3=1$ $x_4=3$ $x_5=0$ $x_6=4.19$

Sentiment Features

Example: Sentiment Analysis – With Engineered features

It's **hokey**. There are virtually **no** surprises, and the writing is **second-rate**. So why was it so **enjoyable**? For one thing, the cast is **great**. Another **nice** touch is the music **I** was overcome with the urge to get off the couch and start dancing. It sucked **me** in, and it'll do the same to **you**.

$x_1=3$ $x_2=2$ $x_3=1$ $x_4=3$ $x_5=0$ $x_6=4.19$

Sentiment Features

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc	3
x_2	count(negative lexicon) $\in$ doc	2
x_3	$\begin{cases} 1 & \text{if "no" } \in \text{ doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!" } \in \text{ doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(66) = 4.19$

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Classifying sentiment using logistic regression

Suppose $w = [2.5, -5.0, -1.2, 0.5, 2.0, 0.7]$

$$b = 0.1$$

$$\begin{aligned} p(+|x) &= P(Y = 1|x) = \sigma(w \cdot x + b) \\ &= \sigma([2.5, -5.0, -1.2, 0.5, 2.0, 0.7] \cdot [3, 2, 1, 3, 0, 4.19] + 0.1) \\ &= \sigma(.833) \\ &= 0.70 \end{aligned}$$

$$\begin{aligned} p(-|x) &= P(Y = 0|x) = 1 - \sigma(w \cdot x + b) \\ &= 0.30 \end{aligned}$$



Logistic Regression – Inference & Interpretation

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

Assume : $0.4 + 0.3CGPA - 0.45IQ$

Predict the Job offered for a candidate : (5, 6)
 $h(x) = 0.31$
 Y-Predicted = 0 / No

Note :
 The exponential function of the regression coefficient ($e^{w \cdot CGPA}$) is the odds ratio associated with a one-unit increase in the cgpa.
 + The odd of being offered with job increase by a factor of 1.35 for every unit increase in the CGPA
`[np.exp(model.params)]`

Logistic Regression – Fit a Model – Apply Gradient Descent

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

$$\begin{aligned} \theta_0 &:= \theta_0 - 0.3 \frac{1}{6} \sum_{i=1}^6 (h_{\theta}(x^{(i)}) - y^{(i)}) (1) \\ \theta_{CGPA} &:= \theta_{CGPA} - 0.3 \frac{1}{6} \sum_{i=1}^6 (h_{\theta}(x^{(i)}) - y^{(i)}) x_{CGPA}^{(i)} \\ \theta_{IQ} &:= \theta_{IQ} - 0.3 \frac{1}{6} \sum_{i=1}^6 (h_{\theta}(x^{(i)}) - y^{(i)}) x_{IQ}^{(i)} \end{aligned}$$

Hyper parameters:
 Learning Rate = 0.3
 Initial Weights = (0.5, 0.5, 0.5)
 Regularization Constant = 0
 $\theta^T X = 0.5 + 0.5 \text{ CGPA} + 0.5 \text{ IQ}$
 $h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$
 $h_{\theta}(x) = \frac{1}{1 + e^{-(0.5 + 0.5 \text{ CGPA} + 0.5 \text{ IQ})}}$

Approx. : New weights
 $\theta_0 = 0.4$
 $\theta_{1=CGPA} = -0.4$
 $\theta_{2=IQ} = -0.6$



Regularization

Note : This topic is already covered in the module 3 and the implementation remains the same.
 Only one points added here w.r.t interpretation for logistic regression



Ways to Control Overfitting – Interpretation of Hyper parameter

- Regularization

$$Loss(S) = \sum_i^n Loss(y_i, \hat{y}_i) + \alpha \sum_j^{\# Weights} |\theta_j|$$



Note:

The hyperparameter controlling the regularization strength of a Scikit-Learn LogisticRegression model is not alpha (as in other linear models), but its inverse: C. The higher the value of C, the less the model is regularized.

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Evaluation of Classifiers

Using Another Example



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Evaluation of Classification Model



Confusion Matrix

Mileage (in kmpl)	Car Price (in cr)
9.8	High
9.12	Low
9.5	High
10	Low
....	...

ACTUAL CLASS	PREDICTED CLASS	
	Class= Low	Class= High
	Class= Low a (TP)	b (FN)
	Class= High c (FP)	d (TN)

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Most effective when the class distribution is relatively balanced

Unseen Data	
Mileage (in kmpl)	Car Price (in cr)
7.5	High
10	Low
....	

Model 1

$$\text{CarPrice} = \frac{1}{1+e^{-8.5 + 0.5 \text{ Mileage} - 1.5 \text{ Mileage}^2}}$$

Accuracy: 99%

Model 2

$$\text{CarPrice} = \frac{1}{1+e^{-5.5 - 1.5 \text{ Mileage}}}$$

Accuracy: 50%

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Evaluation of Classification Model



Confusion Matrix

ACTUAL CLASS	PREDICTED CLASS	
	Class= Low	Class= High
	Class= Low 0 (TP)	10 (FN)
	Class= High 0 (FP)	990 (TN)

If a model predicts everything to be class NO, accuracy is 990/1000 = 99 %. This is **misleading** because this trivial model does not detect any class YES example. Detecting the **rare class** is usually more interesting (e.g., frauds, intrusions, defects, etc)

Model 1

$$\text{CarPrice} = \frac{1}{1+e^{-8.5 + 0.5 \text{ Mileage} - 1.5 \text{ Mileage}^2}}$$

Accuracy: 99%

ACTUAL CLASS	PREDICTED CLASS	
	Class= Low	Class= High
	Class= Low 10 (TP)	0 (FN)
	Class= High 500 (FP)	490 (TN)

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Which model is better?

Model 2

$$\text{CarPrice} = \frac{1}{1+e^{-5.5 - 1.5 \text{ Mileage}}}$$

Accuracy: 50%

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Confusion Matrix

ACTUAL CLASS	PREDICTED CLASS	
	Class=Low	Class=High
	Class=Low a (TP) c (FP)	Class=High b (FN) d (TN)

The F-score (also known as the F1 score or F-measure, combines precision and recall into a single score. F1-score is a better metric when there are imbalanced classes (More on this in upcoming slides)
F-score
= 2 * (precision * recall) / (precision + recall)

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN}$$
$$ErrorRate = 1 - accuracy$$
$$Precision = Positive Predictive Value = \frac{TP}{TP + FP}$$
$$Recall = Sensitivity = TP Rate = \frac{TP}{TP + FN}$$
$$Specificity = TN Rate = \frac{TN}{TN + FP}$$
$$FP Rate = \alpha = \frac{FP}{TN + FP} = 1 - specificity$$
$$FN Rate = \beta = \frac{FN}{FN + TP} = 1 - sensitivity$$
$$Power = sensitivity = 1 - \beta$$

Which Classifier is better?



Low Skew Case

T1	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 50	Class=No 50
	Class=No 1	Class=No 99

Precision (p) = 0.98
TPR = Recall (r) = 0.5
FPR = 0.01
TPR/FPR = 50
F – measure = 0.66

T2	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 99	Class=No 1
	Class=No 10	Class=No 90

Precision (p) = 0.9
TPR = Recall (r) = 0.99
FPR = 0.1
TPR/FPR = 9.9
F – measure = 0.94

T3	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 99	Class=No 1
	Class=No 1	Class=No 99

Precision (p) = 0.99
TPR = Recall (r) = 0.99
FPR = 0.01
TPR/FPR = 99
F – measure = 0.99

Which Classifier is better?



Medium Skew Case

T1	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 50	Class=No 50
	Class=No 10	Class=No 990

Precision (p) = 0.83
TPR = Recall (r) = 0.5
FPR = 0.01
TPR/FPR = 50
F – measure = 0.62

T2	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 99	Class=No 1
	Class=No 100	Class=No 900

Precision (p) = 0.5
TPR = Recall (r) = 0.99
FPR = 0.1
TPR/FPR = 9.9
F – measure = 0.66

T3	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 99	Class=No 1
	Class=No 10	Class=No 990

Precision (p) = 0.9
TPR = Recall (r) = 0.99
FPR = 0.01
TPR/FPR = 99
F – measure = 0.94

Which Classifier is better?



High Skew Case

T1	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 50	Class=No 50
	Class=No 100	Class=No 9900

Precision (p) = 0.3
TPR = Recall (r) = 0.5
FPR = 0.01
TPR/FPR = 50
F – measure = 0.375

T2	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 99	Class=No 1
	Class=No 1000	Class=No 9000

Precision (p) = 0.09
TPR = Recall (r) = 0.99
FPR = 0.1
TPR/FPR = 9.9
F – measure = 0.165

T3	PREDICTED CLASS	
ACTUAL CLASS	Class=Yes	Class=No
	Class=Yes 99	Class=No 1
	Class=No 100	Class=No 9900

Precision (p) = 0.5
TPR = Recall (r) = 0.99
FPR = 0.01
TPR/FPR = 99
F – measure = 0.66

Which Model should you use?

	False Positive Rate	False Negative Rate
Model 1	41%	3%
Model 2	5%	25%

Mistakes have different costs:

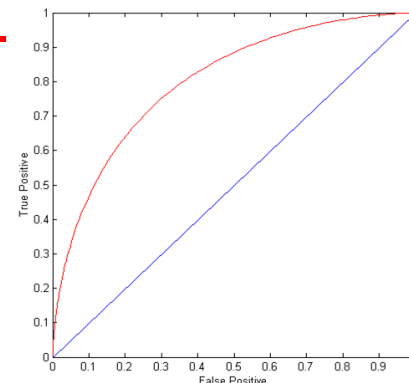
- Disease Screening – LOW FN Rate
- Spam filtering – LOW FP Rate

Conservative vs Aggressive settings:

- The same application might need multiple tradeoffs

ROC (Receiver Operating Characteristic)

- A graphical approach for displaying trade-off between detection rate and false alarm rate
- Developed in 1950s for signal detection theory to analyze noisy signals
- ROC curve plots TPR against FPR
 - Performance of a model represented as a point in an ROC curve

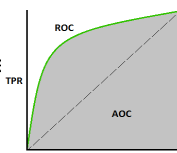


(TPR, FPR):

- (0,0): declare everything to be negative class
- (1,1): declare everything to be positive class
- (1,0): ideal
- Diagonal line:
 - Random guessing
 - Below diagonal line:
 - prediction is opposite of the true class

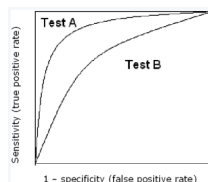
Receiver Operating Characteristic (ROC) Curve

- ROC curve plots TPR and FPR, to graphically represent their trade-off
- AUC represents degree or measure of separability. It tells how much model is capable of distinguishing between classes.
 - Higher the AUC, better the model is at predicting
- Area Under Curve (AUC) of ROC – evaluates model performance on average
 - AUC of ROC = 1, for a perfect model
 - AUC of ROC = 0.5, if the model is random
- For model comparison, AUC of ROC should be larger for the model to be superior or better performing



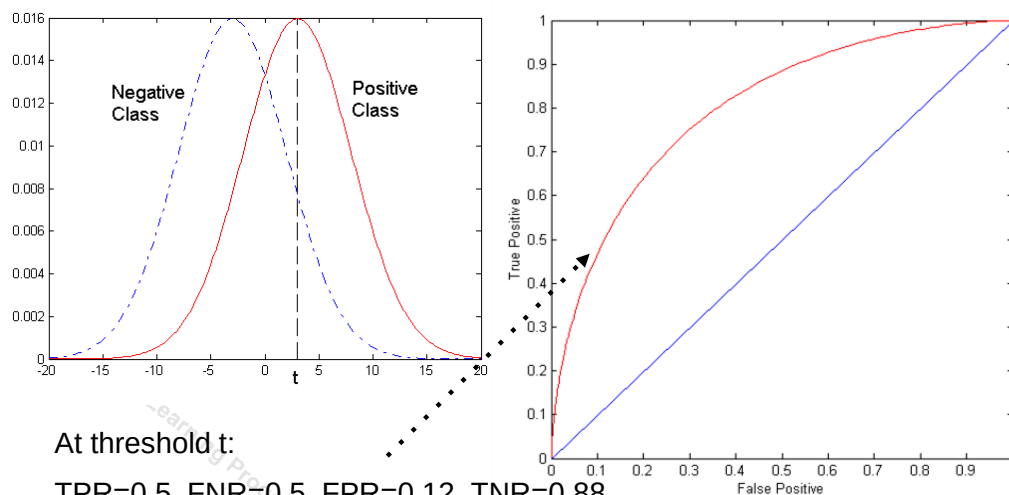
Usage

- Threshold selection
- Performance assessment
- Classifier comparison



ROC Curve Example

- 1-dimensional data set containing 2 classes (positive and negative)
- Any points located at $x > t$ is classified as positive



At threshold t:

TPR=0.5, FNR=0.5, FPR=0.12, TNR=0.88

How to Construct an ROC curve

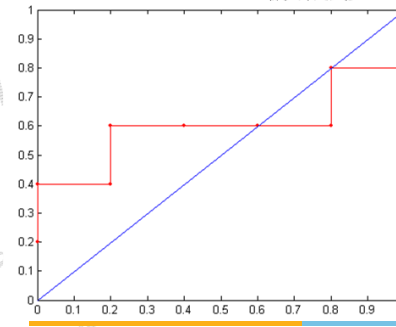
Instance	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	+
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

- Use a classifier that produces a continuous-valued score for each instance
 - The more likely it is for the instance to be in the + class, the higher the score
- Sort the instances in decreasing order according to the score
- Apply a threshold at each unique value of the score
- Count the number of TP, FP, TN, FN at each threshold
 - $TPR = TP/(TP+FN)$
 - $FPR = FP/(FP + TN)$

How to construct an ROC curve

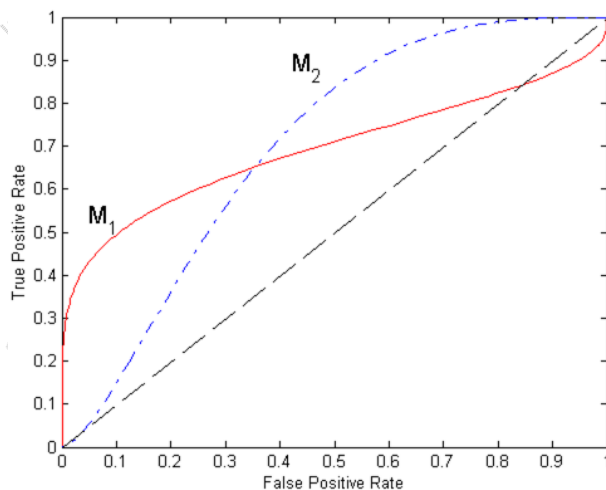
Class	+	-	+	-	-	-	+	-	+	+	
Threshold >=	0.25	0.43	0.53	0.76	0.85	0.85	0.85	0.87	0.93	0.95	1.00
TP	5	4	4	3	3	3	3	2	2	1	0
FP	5	5	4	4	3	2	1	1	0	0	0
TN	0	0	1	1	2	3	4	4	5	5	5
FN	0	1	1	2	2	2	2	3	3	4	5
TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0
FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0

ROC Curve:



Insta nce	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	+
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

Using ROC for Model Comparison

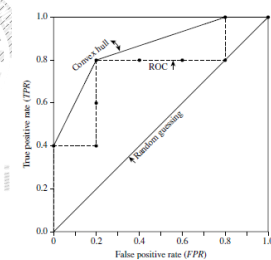


- No model consistently outperforms the other
 - M_1 is better for small FPR
 - M_2 is better for large FPR
- Area Under the ROC curve (AUC)
 - Ideal:
 - Area = 1
 - Random guess:
 - Area = 0.5

Example

The table below shows the probability value (column 3) returned by a probabilistic classifier for each of the 10 tuples in a test set, sorted by decreasing probability order. The corresponding ROC is given on right hand side.

Tuple #	Class	Prob.	TP	FP	TN	FN	TPR	FPR
1	P	0.90	1	0	5	4	0.2	0
2	P	0.80	2	0	5	3	0.4	0
3	N	0.70	2	1	4	3	0.4	0.2
4	P	0.60	3	1	4	2	0.6	0.2
5	P	0.55	4	1	4	1	0.8	0.2
6	N	0.54	4	2	3	1	0.8	0.4
7	N	0.53	4	3	2	1	0.8	0.6
8	N	0.51	4	4	1	1	0.8	0.8
9	P	0.50	5	4	0	1	1.0	0.8
10	N	0.40	5	5	0	0	1.0	1.0



Common Issues in Classifiers

Sigmoid Function

- The main class of interest is **rare**.
- The data set distribution reflects a significant majority of the negative class (Eg., Job Offered = Yes/1) and a minority positive class (Eg., Job Offered = No/0)
- For Another Example,
 - fraud detection applications, the class of interest (or positive class) is “*fraud*,”
 - medical tests, there may be a rare class, such as “*cancer*”
- **Accuracy might not be a good option** for measuring performance in case of class imbalance problem

Solution to Class Imbalance

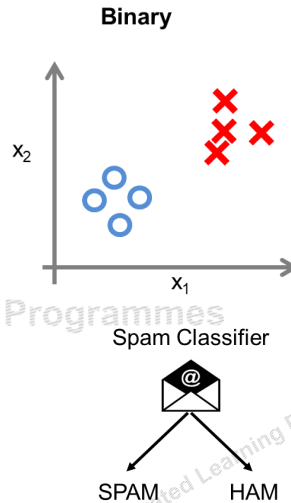
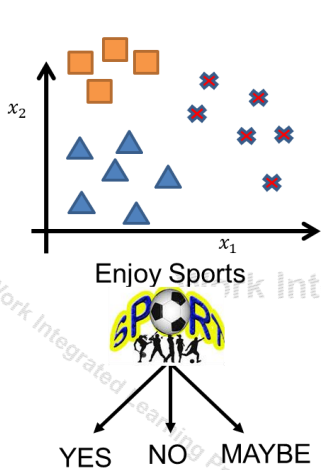
- Generate Synthetic Samples
- New samples based on the distances between the point and its nearest neighbors E.g. Synthetic Minority Oversampling Technique, or **SMOTE class in sklearn**
- Change the performance metric : Use Recall, Precision or ROC curves instead of accuracy
- Try different algorithms : Some algorithms as Support Vector Machines and Tree-Based algorithms may work better with imbalanced classes. We will discuss these post mid term

Types of Classification Based on the Output Labels

Types of Classification

Output Labels

- Target Concept

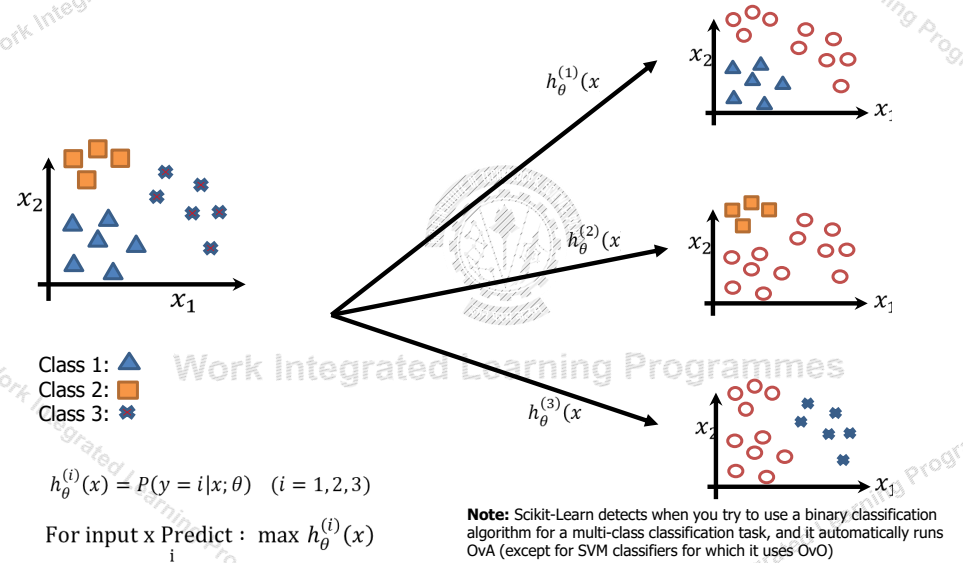


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Prediction – Multi class Classification

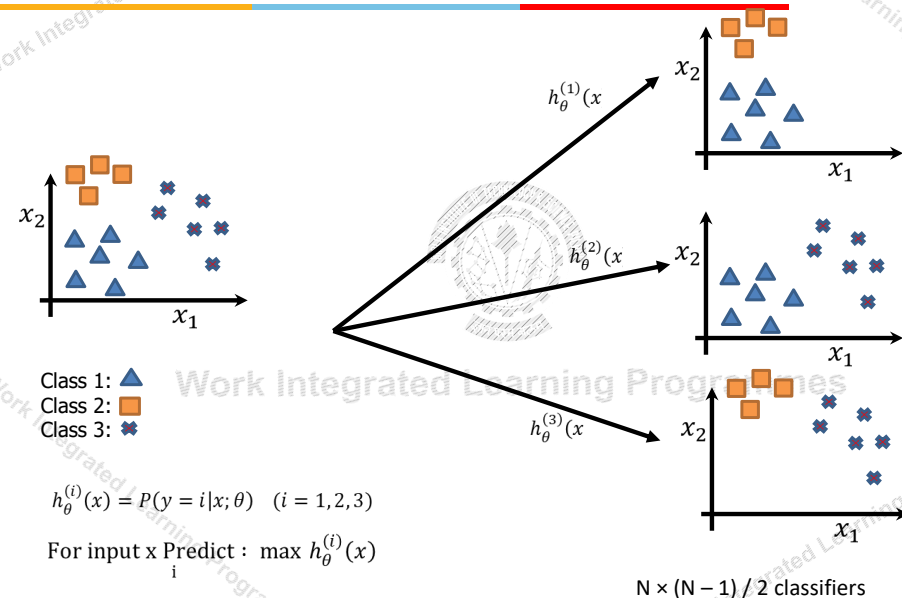
One Vs All Strategy(one-vs-rest)



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Prediction – Multi class Classification

One Vs One Strategy



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Logistic regression (Classification)- Summary

Model

$$h_{\theta}(x) = P(Y = 1 | X_1, X_2, \dots, X_n) = \frac{1}{1 + e^{-\theta^T x}}$$

Cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \quad \text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

Learning

Gradient descent: Repeat $\{\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}\}$

Inference

$$\hat{y} = h_{\theta}(x^{\text{test}}) = \frac{1}{1 + e^{-\theta^T x^{\text{test}}}}$$

Note:

- $\sigma(t) < 0.5$ when $t < 0$, and $\sigma(t) \geq 0.5$ when $t \geq 0$, so a Logistic model predicts 1 if $x^T \theta$ is positive, and 0 if it is negative
- $\text{logit}(p) = \log(p / (1 - p))$, is the inverse of the logistic function. Indeed, if you compute the logit of the estimated probability p , you will find that the result is t . The logit is also called the log-odds

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